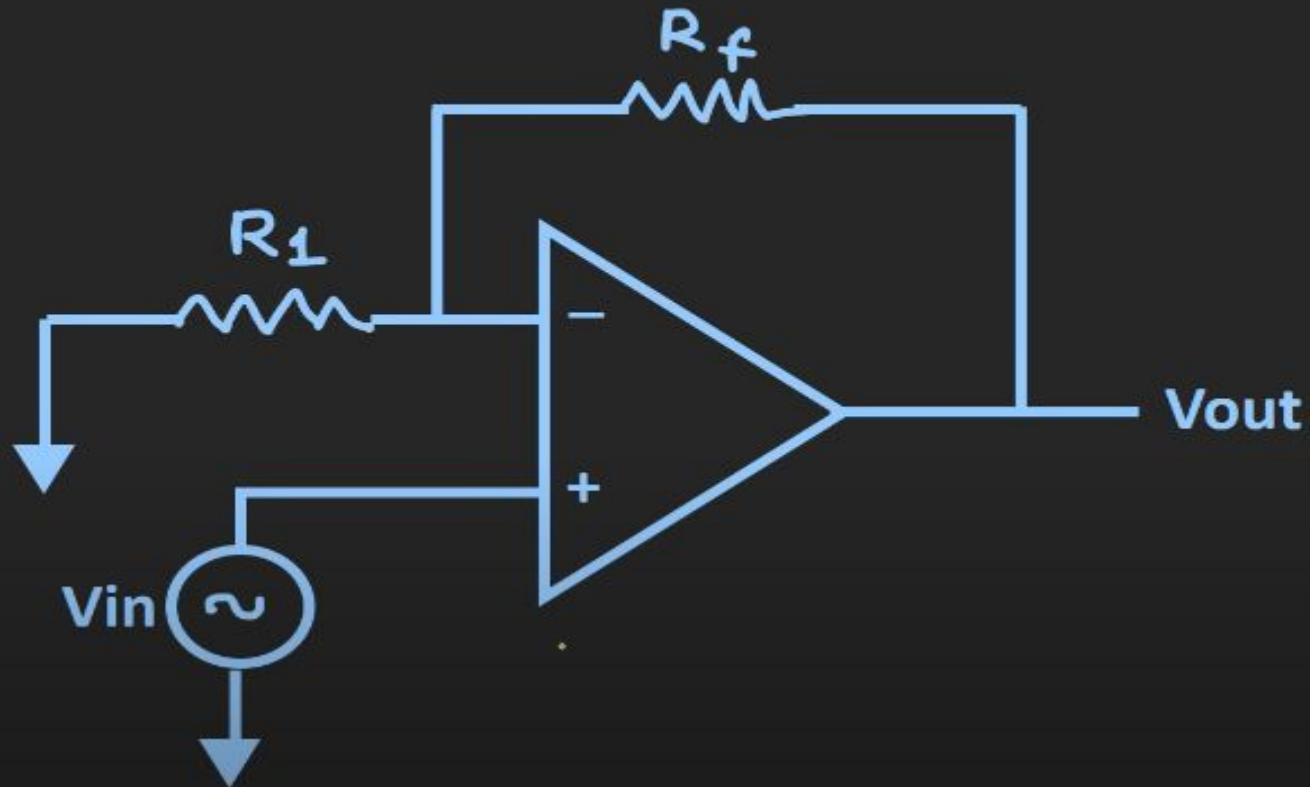
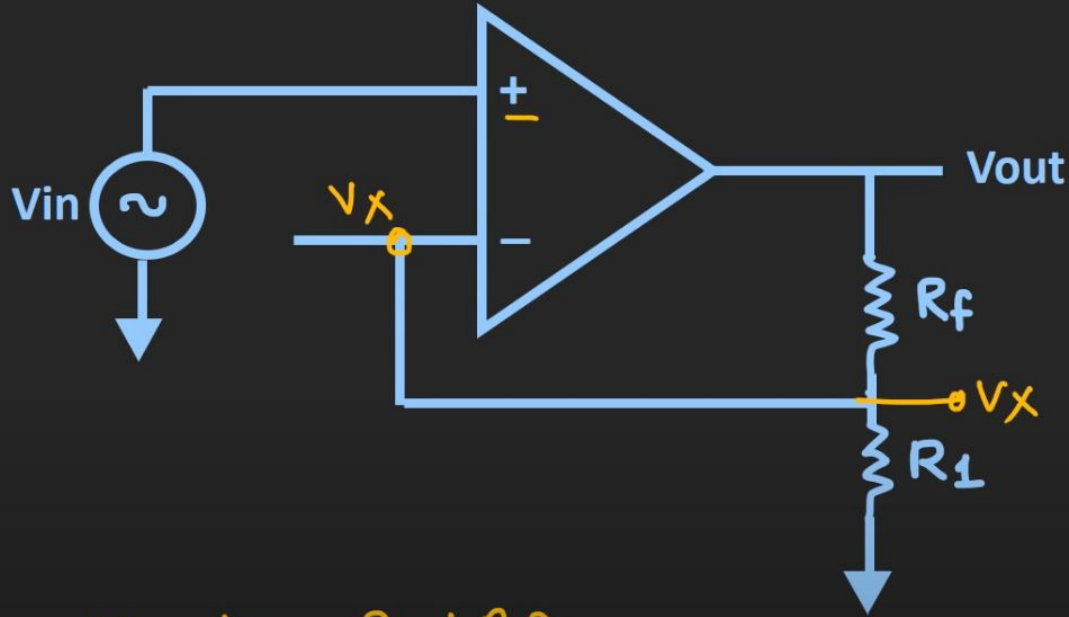


Non Inverting Amplifier

Non-Inverting Operational Amplifier





$$V_x = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$V^+ = V^- \quad (\because \text{virtual short})$$

$$V^+ = V_{in}$$

$$V^- = V_{in} = V_x$$

$$\frac{V_{out}}{V_{in}} = \frac{R_1 + R_f}{R_1} = \left(1 + \frac{R_f}{R_1}\right) = A_{CL}$$

$$V_{in} = \frac{R_1}{R_1 + R_f} \times V_{out}$$

The closed loop gain of the circuit is,

$$A = \left(1 + \frac{R_f}{R_1}\right)$$

This term does not contain any negative part. Hence, it proves that the input signal to the circuit gets amplified without changing its polarity at the output.

From the expression of **voltage** gain of an **non inverting op amp**, it is clear that, the gain will be unity when $R_f = 0$ or $R_1 \rightarrow \infty$.

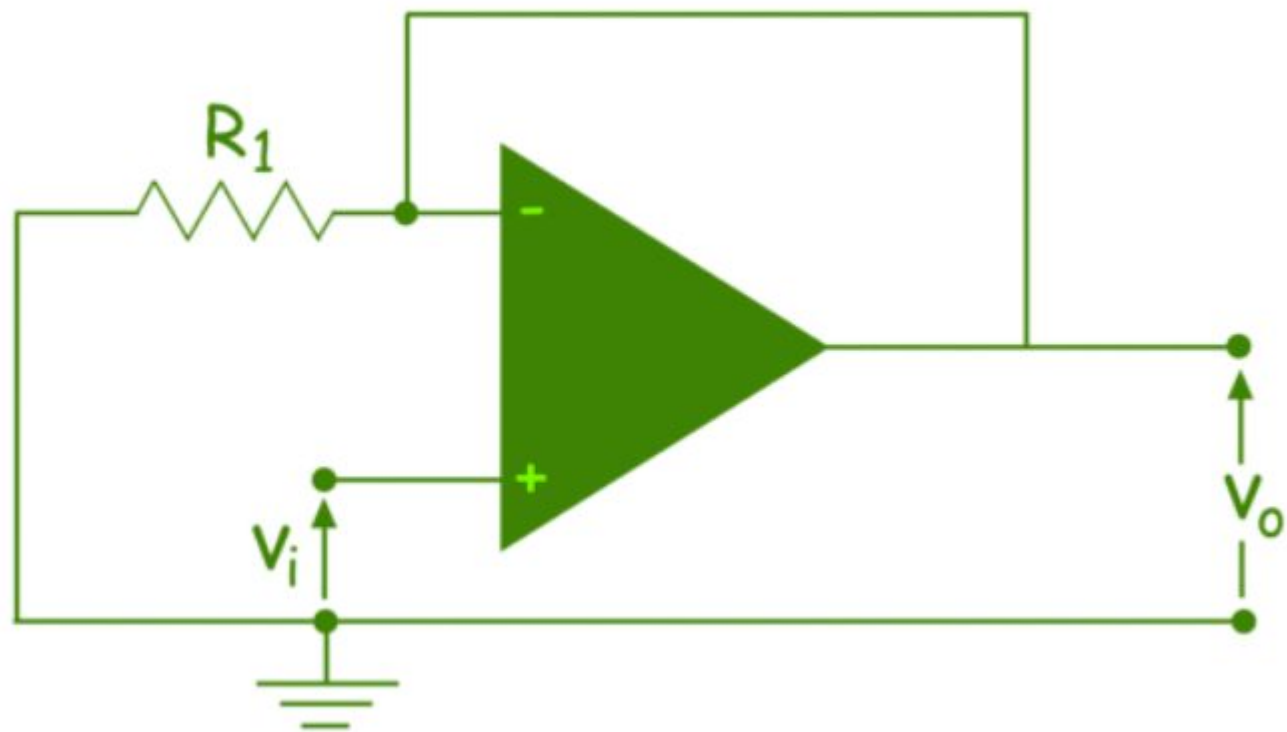
When, $R_f = 0$

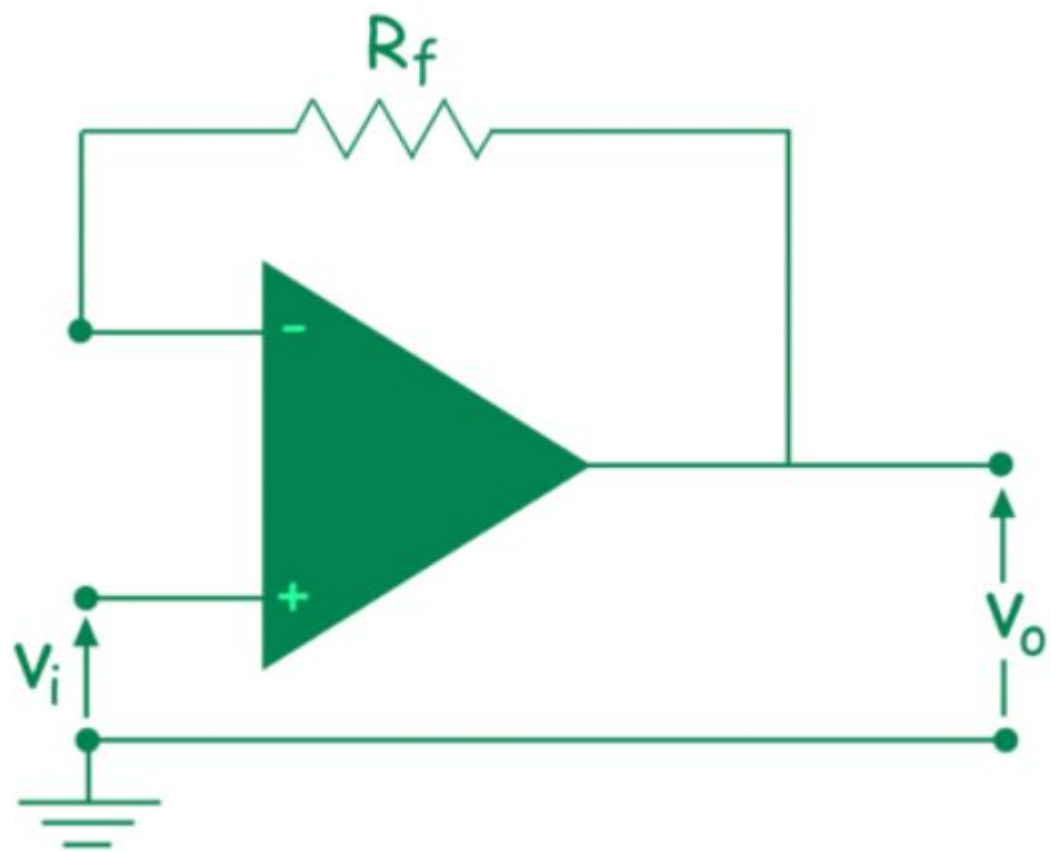
$$A = \left(1 + \frac{R_f}{R_1}\right) = \left(1 + \frac{0}{R_1}\right) = 1$$

When, $R_1 \rightarrow \infty$

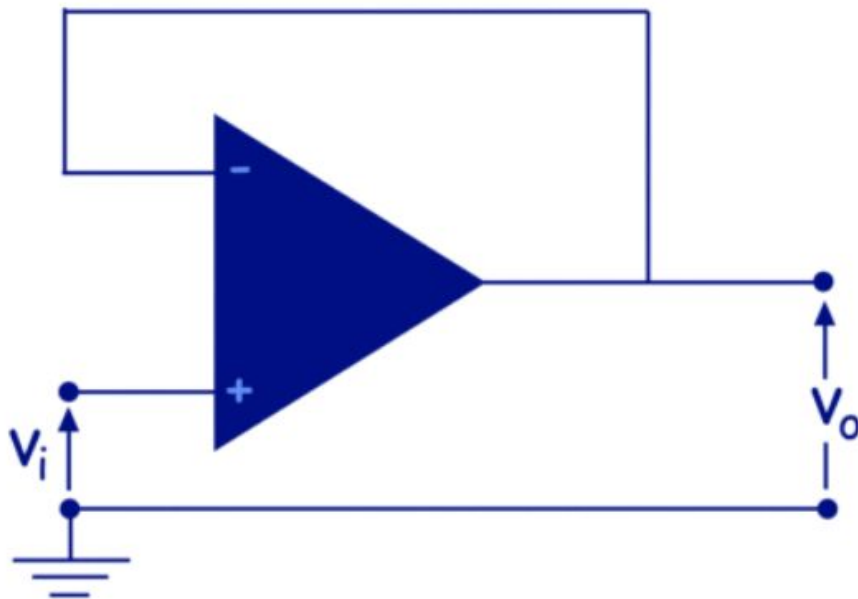
$$A = \left(1 + \frac{R_f}{R_1}\right) = \left(1 + \frac{R_f}{\infty}\right) = 1$$

So, if we short circuit the feedback path and/or open the external **resistance** of the inverting pin, the gain of the circuit becomes 1.





Voltage follower

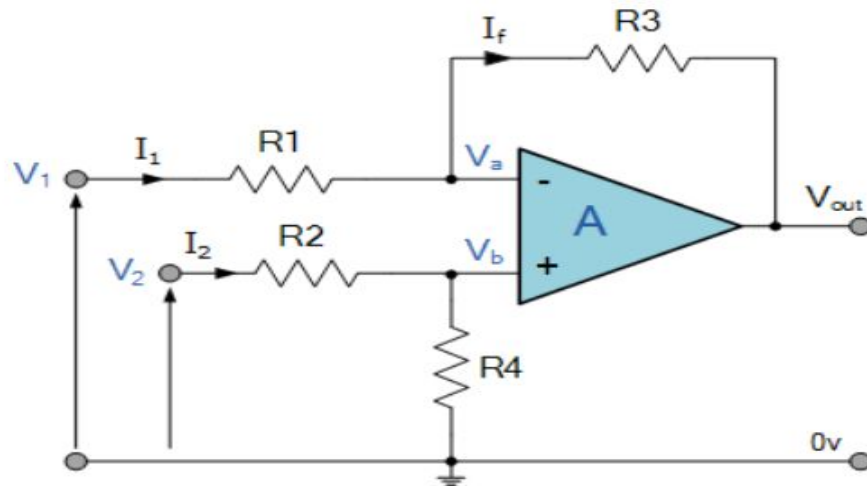


Thus far we have used only one of the operational amplifiers inputs to connect to the amplifier, using either the “inverting” or the “non-inverting” input terminal to amplify a single input signal with the other input being connected to ground.

But as a standard operational amplifier has two inputs, inverting and no-inverting, we can also connect signals to both of these inputs at the same time producing another common type of operational amplifier circuit called a **Differential Amplifier**.

Then *differential amplifiers* amplify the difference between two voltages making this type of operational amplifier circuit a **Subtractor** unlike a summing amplifier which adds or sums together the input voltages.

Differential Amplifier



By connecting each input in turn to 0v ground we can use superposition to solve for the output voltage V_{out} . Then the transfer function for a **Differential Amplifier** circuit is given as:

$$I_1 = \frac{V_1 - V_a}{R_1}, \quad I_2 = \frac{V_2 - V_b}{R_2}, \quad I_f = \frac{V_a - (V_{out})}{R_3}$$

Summing point $V_a = V_b$

$$\text{and } V_b = V_2 \left(\frac{R_4}{R_2 + R_4} \right)$$

$$\text{If } V_2 = 0, \text{ then: } V_{\text{out(a)}} = -V_1 \left(\frac{R_3}{R_1} \right)$$

$$\text{If } V_1 = 0, \text{ then: } V_{\text{out(b)}} = V_2 \left(\frac{R_4}{R_2 + R_4} \right) \left(\frac{R_1 + R_3}{R_1} \right)$$

$$V_{\text{out}} = -V_{\text{out(a)}} + V_{\text{out(b)}}$$

$$\therefore V_{\text{out}} = -V_1 \left(\frac{R_3}{R_1} \right) + V_2 \left(\frac{R_4}{R_2 + R_4} \right) \left(\frac{R_1 + R_3}{R_1} \right)$$

When resistors, $R_1 = R_2$ and $R_3 = R_4$ the above transfer function for the differential amplifier can be simplified to the following expression:

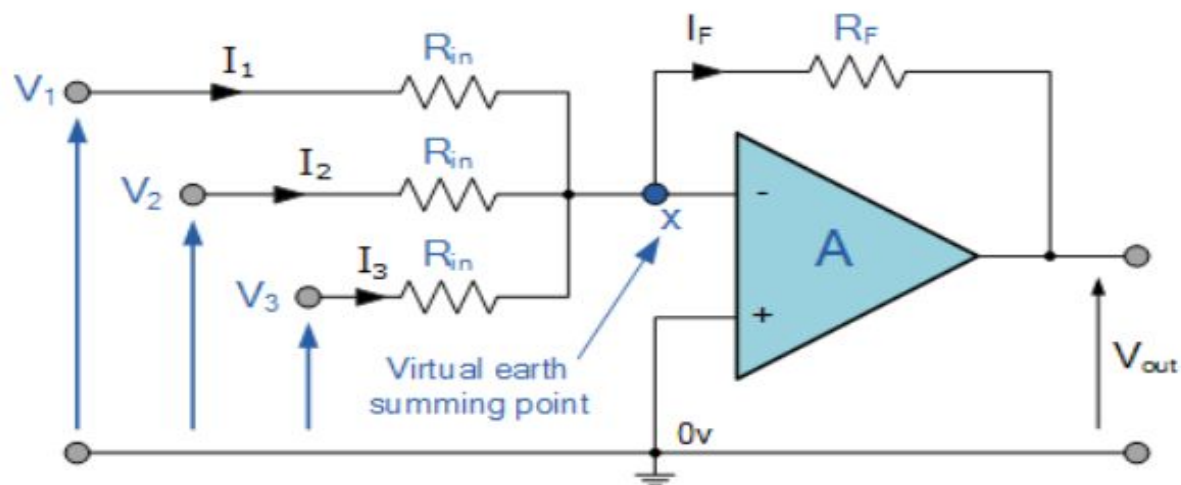
Differential Amplifier Equation

$$V_{OUT} = \frac{R_3}{R_1} (V_2 - V_1)$$

If all the resistors are all of the same ohmic value, that is: $R_1 = R_2 = R_3 = R_4$ then the circuit will become a **Unity Gain Differential Amplifier** and the voltage gain of the amplifier will be exactly one or unity. Then the output expression would simply be $V_{out} = V_2 - V_1$.

Also note that if input V_1 is higher than input V_2 the output voltage sum will be negative, and if V_2 is higher than V_1 , the output voltage sum will be positive.

Summing Amplifier Circuit



In this simple summing amplifier circuit, the output voltage, (V_{out}) now becomes proportional to the sum of the input voltages, V_1 , V_2 , V_3 , etc. Then we can modify the original equation for the inverting amplifier to take account of these new inputs thus:

$$I_F = I_1 + I_2 + I_3 = - \left[\frac{V_1}{R_{in}} + \frac{V_2}{R_{in}} + \frac{V_3}{R_{in}} \right]$$

Inverting Equation: $V_{out} = -\frac{R_f}{R_{in}} \times V_{in}$

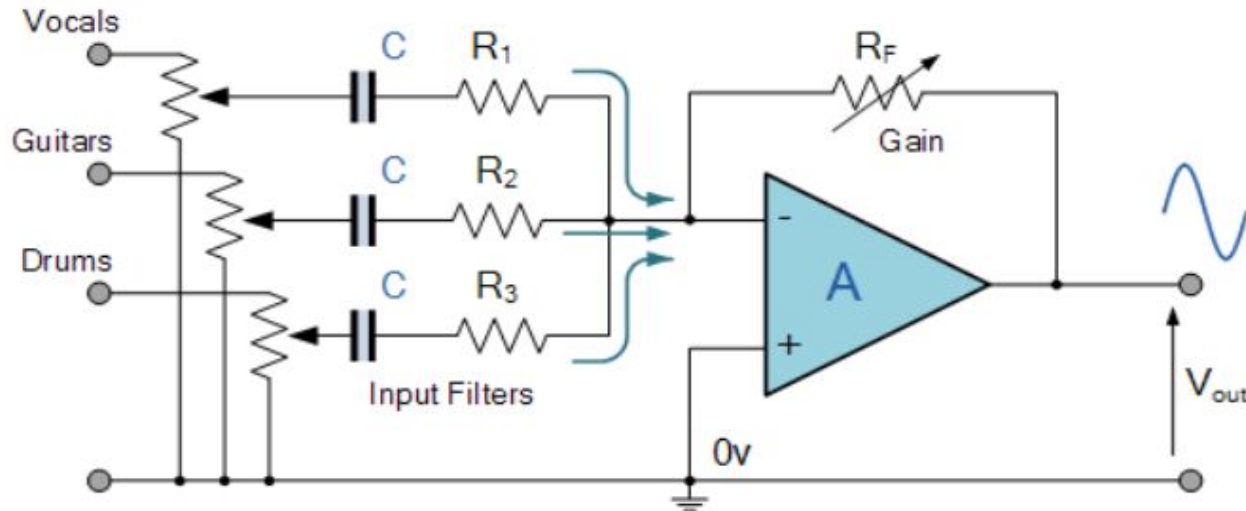
then, $-V_{out} = \left[\frac{R_F}{R_{in}} V_1 + \frac{R_F}{R_{in}} V_2 + \frac{R_F}{R_{in}} V_3 \right]$

However, if all the input impedances, (R_{IN}) are equal in value, we can simplify the above equation to give an output voltage of:

Summing Amplifier Equation

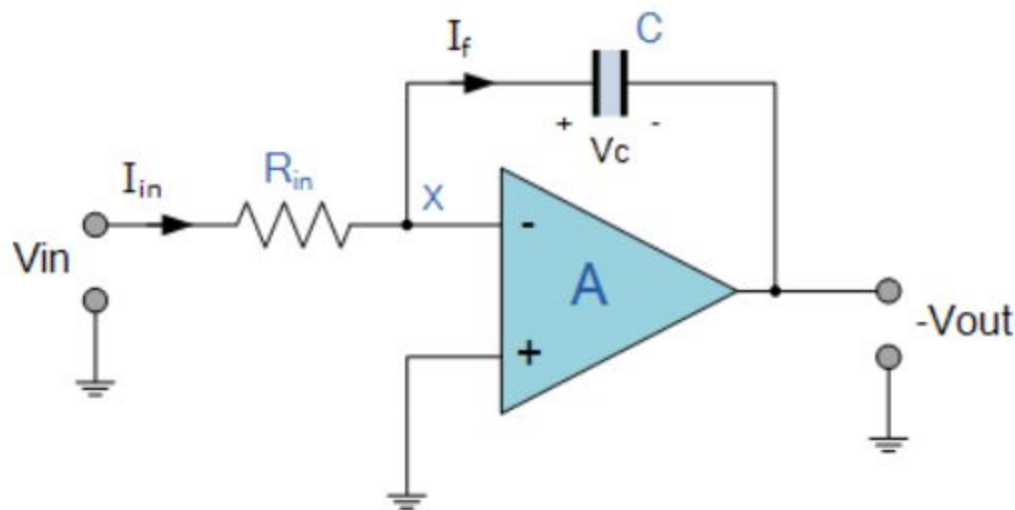
$$-V_{out} = \frac{R_F}{R_{IN}} \left(V_1 + V_2 + V_3 \dots \text{etc} \right)$$

Summing Amplifier Audio Mixer



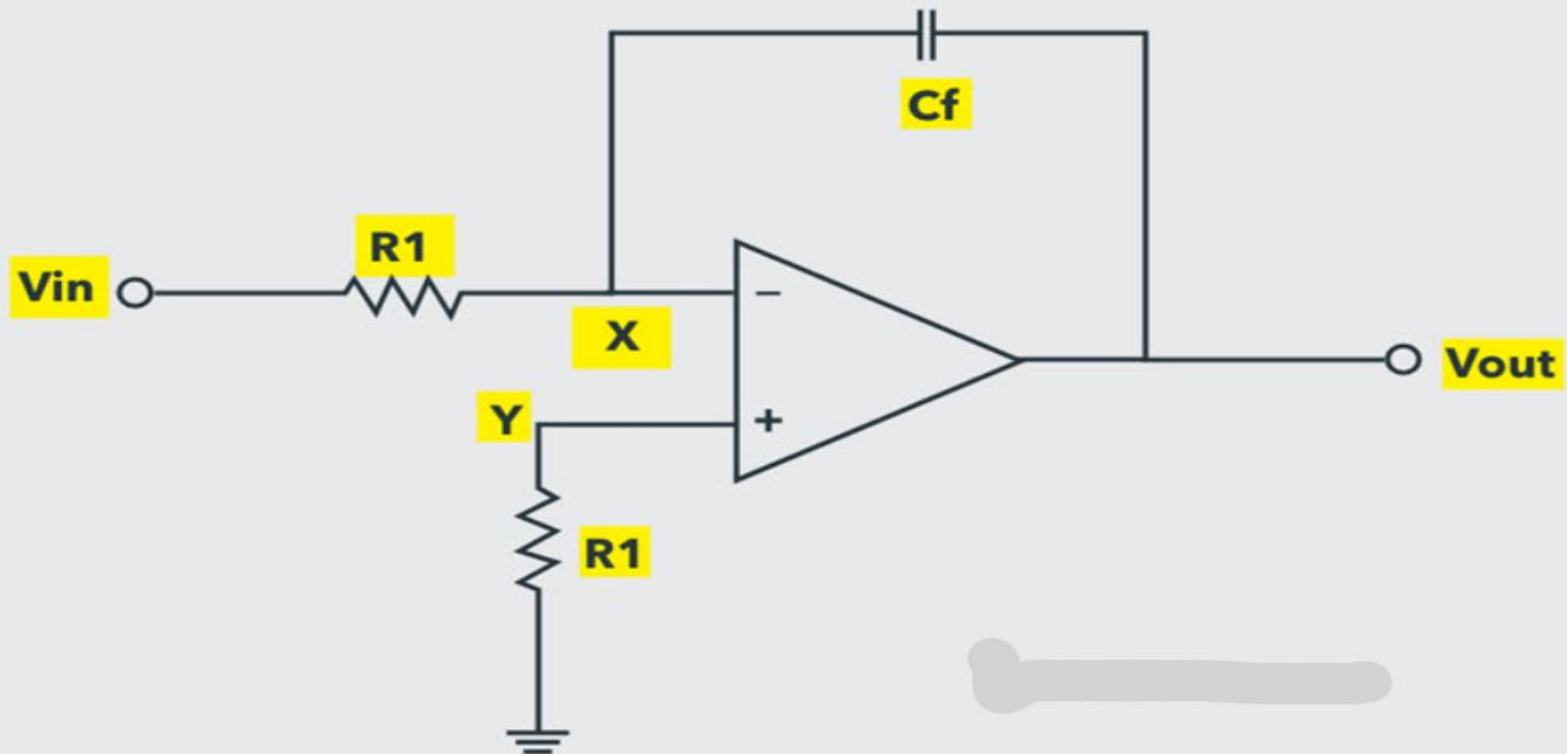
Another useful application of a **Summing Amplifier** is as a weighted sum digital-to-analogue converter, (DAC). If the input resistors, R_{IN} of the summing amplifier double in value for each input, for example, $1k\Omega$, $2k\Omega$, $4k\Omega$, $8k\Omega$, $16k\Omega$, etc, then a digital logical voltage, either a logic level "0" or a logic level "1" on these inputs will produce an output which is the weighted sum of the digital inputs. Consider the circuit below.

Op-amp Integrator Circuit



As its name implies, the **Op-amp Integrator** is an operational amplifier circuit that performs the mathematical operation of **Integration**, that is we can cause the output to respond to changes in the input voltage over time as the op-amp integrator produces an *output voltage which is proportional to the integral of the input voltage*.

Ideal Op-Amp Integrator Circuit



Output Voltage Calculation

From the circuit, it is seen that node Y is grounded through a compensating resistor R_1 . Node X will also be at ground potential, due to the virtual ground.

$$V_X = V_Y = 0$$

Since the input current to an op-amp is ideally zero, the current flowing through the input resistor, due to V_{in} , also flows through the capacitor C_f .

From the input side, the current I is given as,

$$I = (V_{IN} - V_X) / R_1 = V_{IN} / R_1$$

From the output side, the current I is given as,

$$I = C_f [d(V_X - V_{OUT})/dt] = -C_f [d(V_{OUT})/dt]$$

Equating the above two equations of I, we get,

$$[V_{IN} / R_1] = - C_f [d(V_{OUT})/dt]$$

Integrating both the sides of the above equation,

$$\int_0^t \frac{V_{in}}{R_1} . dt = - C_f \int_0^t d \frac{V_{out}}{dt} dt$$

$$\int_0^t \frac{V_{in}}{R_1} . dt = -C_f . V_{out}$$

$$\text{Therefore, } V_{out} = - \frac{1}{R_1 . C_f} \int_0^t V_{in} . dt$$

In the above equation, the output is $\{-1/(R_1 * C_f)\}$ times the integral of the input voltage, where the term $(R_1 * C_f)$ is known as the time constant of the integrator.

The negative sign indicates that there is a phase shift of 180° between input and output, because the input is provided to the inverting input terminal of the op-amp.

Ideal Op-Amp Differentiator Circuit

An op-amp differentiating amplifier uses a capacitor in series with the input voltage source, as shown in the figure below.

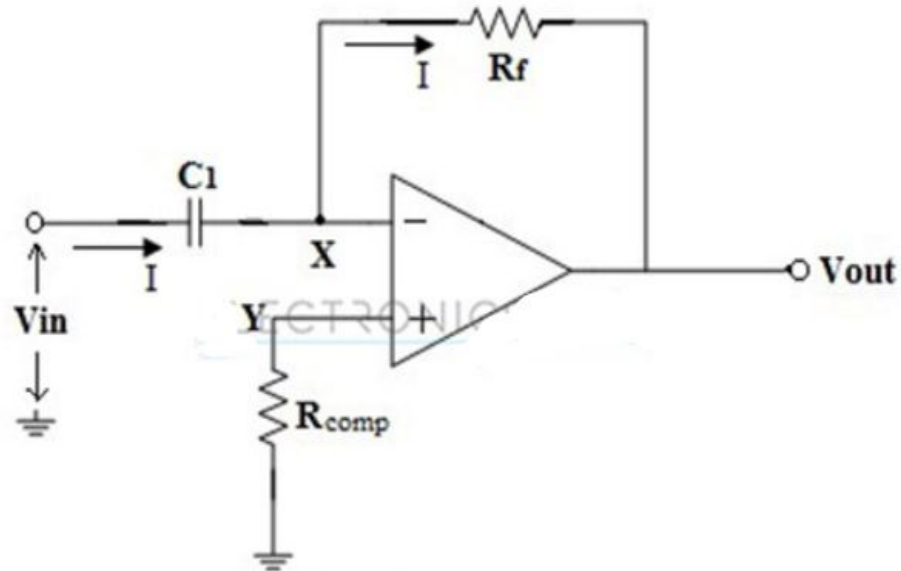


Fig. An Ideal Op-amp Differentiator

Thus, the circuit behaves like a voltage follower.

When the input is a positive-going voltage, a current I flows into the capacitor C_1 , as shown in the figure. Since the current flowing into the op-amp's internal circuit is zero, effectively all of the current I flows through the resistor R_f . The output voltage is,

$$V_{\text{out}} = - (I * R_f)$$

Here, this output voltage is directly proportional to the rate of change of the input voltage.

From the figure, node 'X' is virtually grounded and node 'Y' is also at ground potential i.e., $V_X = V_Y = 0$.

From the input side, the current I can be given as:

$$I = C_1 \{d(V_{\text{in}} - V_X) / dt\} = C_1 \{d(V_{\text{in}}) / dt\}$$

From the output side, the current I is given as:

$$I = -(V_{\text{out}} - V_X) / R_f = -V_{\text{out}} / R_f$$

Equating the above two equations of current we get:

$$C_1 \{d(V_{\text{in}}) / dt\} = -V_{\text{out}} / R_f$$

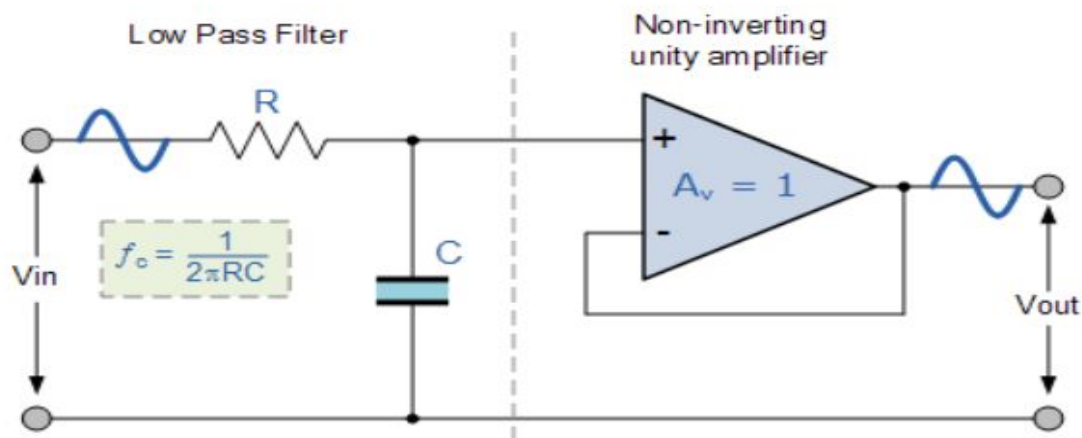
$$V_{\text{out}} = -C_1 R_f \{d(V_{\text{in}}) / dt\}$$

Above equation indicates that the output is $C_1 R_f$ times the differentiation of the input voltage. The product $C_1 R_f$ is called as the RC time constant of the differentiator circuit. The negative sign indicates the output is out of phase by 180° with respect to the input.

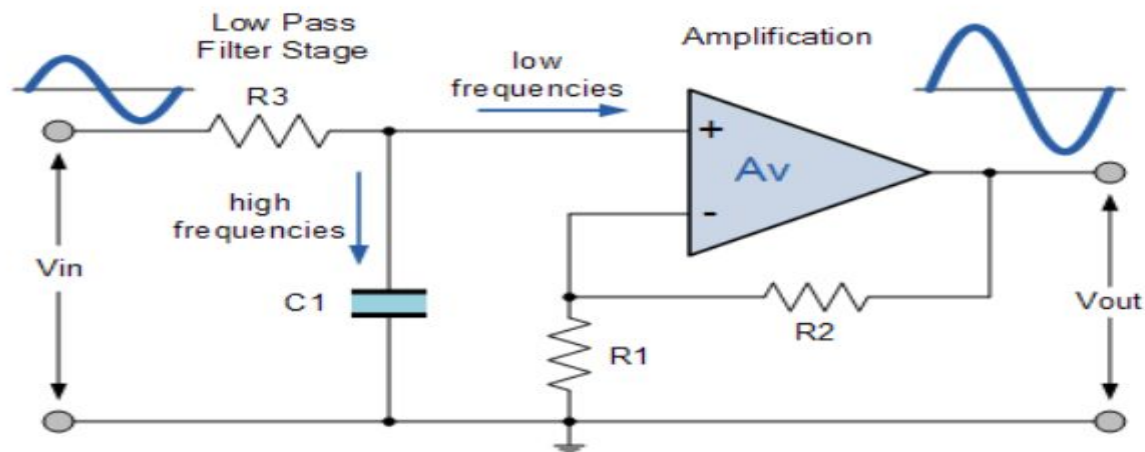
Active Low Pass Filter

The most common and easily understood active filter is the **Active Low Pass Filter**. Its principle of operation and frequency response is exactly the same as those for the previously seen passive filter, the only difference this time is that it uses an op-amp for amplification and gain control. The simplest form of a low pass active filter is to connect an inverting or non-inverting amplifier, the same as those discussed in the Op-amp tutorial, to the basic RC low pass filter circuit as shown.

First Order Low Pass Filter



Active Low Pass Filter with Amplification



The frequency response of the circuit will be the same as that for the passive RC filter, except that the amplitude of the output is increased by the pass band gain, A_F of the amplifier. For a non-inverting amplifier circuit, the magnitude of the voltage gain for the filter is given as a function of the feedback resistor (R_2) divided by its corresponding input resistor (R_1) value and is given as:

$$\text{DC gain} = \left(1 + \frac{R_2}{R_1} \right)$$

Therefore, the gain of an active low pass filter as a function of frequency will be:

Gain of a first-order low pass filter

$$\text{Voltage Gain, } (A_v) = \frac{V_{out}}{V_{in}} = \frac{A_F}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Where:

A_F = the pass band gain of the filter, $(1 + R_2/R_1)$

f = the frequency of the input signal in Hertz, (Hz)

f_c = the cut-off frequency in Hertz, (Hz)

Thus, the operation of a low pass active filter can be verified from the frequency gain equation above as:

- | | |
|--|---|
| 1. At very low frequencies, $f < f_c$ | $\frac{V_{out}}{V_{in}} \cong A_F$ |
| 2. At the cut-off frequency, $f = f_c$ | $\frac{V_{out}}{V_{in}} = \frac{A_F}{\sqrt{2}} = 0.707 A_F$ |
| 3. At very high frequencies, $f > f_c$ | $\frac{V_{out}}{V_{in}} < A_F$ |

Thus, the **Active Low Pass Filter** has a constant gain A_F from 0Hz to the high frequency cut-off point, f_C . At f_C the gain is $0.707A_F$, and after f_C it decreases at a constant rate as the frequency increases. That is, when the frequency is increased tenfold (one decade), the voltage gain is divided by 10.

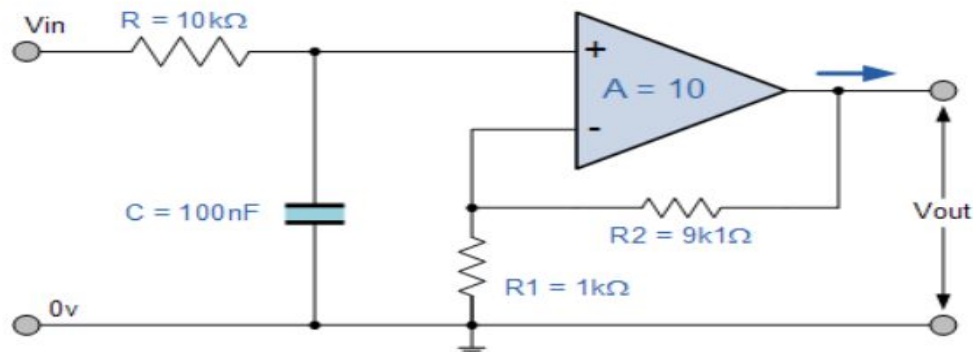
In other words, the gain decreases 20dB ($= 20 \cdot \log(10)$) each time the frequency is increased by 10. When dealing with filter circuits the magnitude of the pass band gain of the circuit is generally expressed in *decibels* or *dB* as a function of the voltage gain, and this is defined as:

Magnitude of Voltage Gain in (dB)

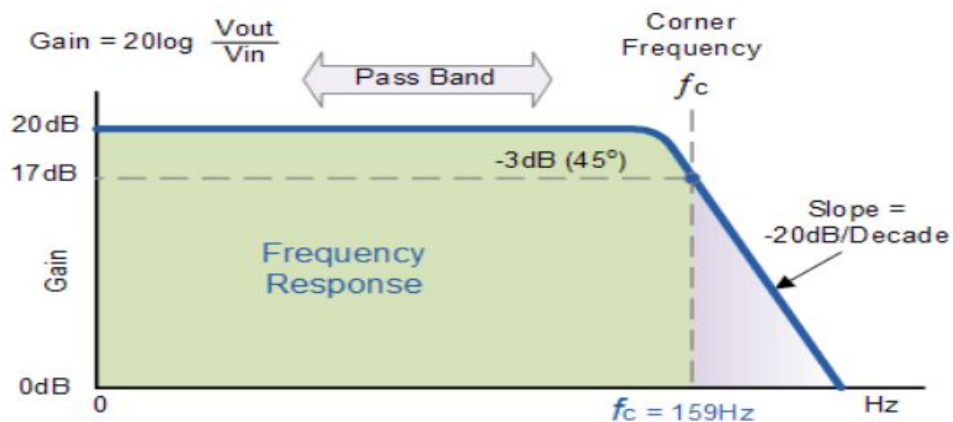
$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{V_{\text{out}}}{V_{\text{in}}} \right)$$

$$\therefore -3\text{dB} = 20 \log_{10} \left(0.707 \frac{V_{\text{out}}}{V_{\text{in}}} \right)$$

Low Pass Filter Circuit



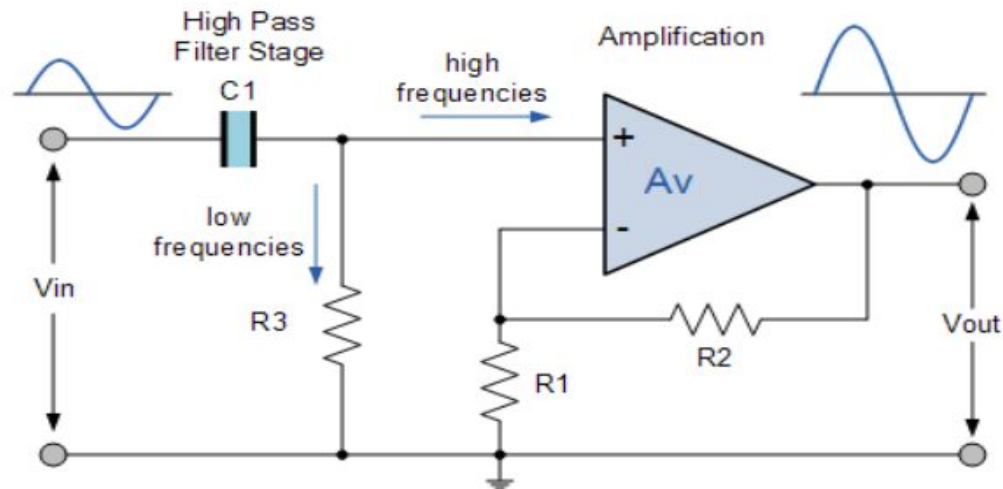
Frequency Response Curve



Active High Pass Filter

A first-order (single-pole) **Active High Pass Filter** as its name implies, attenuates low frequencies and passes high frequency signals. It consists simply of a passive filter section followed by a non-inverting operational amplifier. The frequency response of the circuit is the same as that of the passive filter, except that the amplitude of the signal is increased by the gain of the amplifier and for a non-inverting amplifier the value of the pass band voltage gain is given as $1 + R2/R1$, the same as for the low pass filter circuit.

Active High Pass Filter with Amplification



This *first-order high pass filter*, consists simply of a passive filter followed by a non-inverting amplifier. The frequency response of the circuit is the same as that of the passive filter, except that the amplitude of the signal is increased by the gain of the amplifier.

For a non-inverting amplifier circuit, the magnitude of the voltage gain for the filter is given as a function of the feedback resistor (R_2) divided by its corresponding input resistor (R_1) value and is given as:

Gain for an Active High Pass Filter

$$\text{Voltage Gain, (A}_v\text{)} = \frac{V_{out}}{V_{in}} = \frac{A_F \left(\frac{f}{f_c} \right)}{\sqrt{1 + \left(\frac{f}{f_c} \right)^2}}$$

Where:

A_F = the Pass band Gain of the filter, ($1 + R_2/R_1$)

f = the Frequency of the Input Signal in Hertz, (Hz)

f_c = the Cut-off Frequency in Hertz, (Hz)

Just like the low pass filter, the operation of a high pass active filter can be verified from the frequency gain equation above as:

1. At very low frequencies, $f < f_C$ $\frac{V_{out}}{V_{in}} < A_F$
2. At the cut-off frequency, $f = f_C$ $\frac{V_{out}}{V_{in}} = \frac{A_F}{\sqrt{2}} = 0.707 A_F$
3. At very high frequencies, $f > f_C$ $\frac{V_{out}}{V_{in}} \equiv A_F$

Then, the **Active High Pass Filter** has a gain A_F that increases from 0Hz to the low frequency cut-off point, f_C at 20dB/decade as the frequency increases. At f_C the gain is $0.707 \cdot A_F$, and after f_C all frequencies are pass band frequencies so the filter has a constant gain A_F with the highest frequency being determined by the closed loop bandwidth of the op-amp.

When dealing with filter circuits the magnitude of the pass band gain of the circuit is generally expressed in *decibels* or *dB* as a function of the voltage gain, and this is defined as:

Magnitude of Voltage Gain in (dB)

$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{V_{out}}{V_{in}} \right)$$

$$\therefore -3\text{dB} = 20 \log_{10} \left(0.707 \frac{V_{out}}{V_{in}} \right)$$